

# Green Synthesis of *Ceiba pentandra* copper nanoparticles and their characterization

Riazunnisa K.<sup>1\*</sup>, Rajesh N.<sup>1</sup>, Chandrasekhar T.<sup>2</sup>, Anu Prasanna V.<sup>3</sup> and Subhosh Chandra M.<sup>4</sup>

1. Department of Biotechnology and Bioinformatics, Yogi Vemana University, Kadapa-516 005, Andhra Pradesh, INDIA

2. Department of Environmental Science, Yogi Vemana University, Kadapa-516005, Andhra Pradesh, INDIA

3. Department of Zoology, Yogi Vemana University, Kadapa-516005, Andhra Pradesh, INDIA

4. Department of Microbiology, Yogi Vemana University, Kadapa-516005, Andhra Pradesh, INDIA

\*khateefriaz@gmail.com; krbtbi@yogivemanauniversity.ac.in

## Abstract

In the present study, *Ceiba pentandra* (CP) aqueous leaf extract was used for the synthesis of copper nanoparticles (CuNPs). Techniques like UV-visible spectrophotometer, SEM, EDX, XRD and *in vitro* antioxidants assay were used to characterize synthesized nanoparticles. UV-Visible spectrophotometer showed absorbance peak at 310 nm. SEM analysis demonstrated that the CuNPs are spindly shaped structures with average size of about 62.63nm.

XRD analysis revealed the face centered cubic like structures of CuNPs. *Ceiba pentandra* copper nanoparticles (CP-CuNPs) exhibited good *in vitro* antioxidant activities i.e. 80 % inhibition of free radicals with DPPH assay, 70% inhibition of H<sub>2</sub>O<sub>2</sub> with hydrogen peroxide scavenging assay and 75% inhibition in phosphomolybdenum assay compared to leaf extract. Hence the above protocol is a humble, fast, environmentally friendly, non-hazardous and an alternative for physical/chemical methods.

**Keywords:** *Ceiba pentandra*, Copper nanoparticles, SEM, EDX, XRD, *In vitro* antioxidant assay.

## Introduction

Nanotechnology is the study of materials at the nano-level. Various metals are existing in the nature, but copper is becoming popular worldwide. Green synthesis of nanoparticles is ecofriendly and is an alternative for physical/chemical methods becoming progressively popular worldwide<sup>26</sup>. Copper is a popular metal due to its specialized properties which include electrical, optical, catalytic, biomedical and biocidal applications against microbes (bacteria, fungi, viruses and algae)<sup>11,22,24,27</sup>. Due to the antibacterial activity of copper<sup>7,8,20</sup>, copper nanoparticles (CuNPs) are gaining popularity and copper containing surfaces are used for sterilization in hospitals<sup>14</sup>. Copper is toxic to microbes and safe to humans, hence used in food packaging and water treatment applications<sup>6,17,19</sup>.

Ion sputtering, chemical reduction, sol gel<sup>1,23,25,28</sup> are the available protocols for copper nanoparticles synthesis, but these are hazardous, require high energy equipment and are difficult. Green synthesis is gaining popularity due to its nonhazardous, ecofriendly nature, cost effective, simple,

robust, usage of less temperature and compatible to medical and food applications<sup>2,10,15</sup>. Microbial synthesis of nanoparticles is slow as compared to the plants<sup>13</sup>. Very recently plant extracts of *Hagenia abyssinica*<sup>18</sup>, *Jatropha curcas*<sup>21</sup>, *Celastrus paniculatus*<sup>4</sup>, *Cocculus hirsutus*<sup>3</sup> have been used for the synthesis of CuNPs.

All parts of *C. pentandra* plant are used as medicine and its flowers and roots are used for various therapeutic purposes for treating a large number of diseases. Different parts of this plant are utilized in traditional medicines like Ayurveda. Leaf extract has various medicinal properties like antidysentric, leucorrhoea, anemia and infertility.

In the present study, leaf extract of *Ceiba pentandra* (commonly known as Burugu), a species of family Bombacaceae, was used for the synthesis and characterization of *Ceiba pentandra* of CuNPs.

## Material and Methods

**Collection of leaves:** *Ceiba pentandra* leaves were collected from the Botanical Garden and taxonomically authenticated by Dept. of Botany, Yogi Vemana University, Kadapa, Andhra Pradesh, India (Fig. 1). Leaves were washed with running tap water for 3-4 times, later washed with distilled water twice and were shade dried. Leaves were grinded, powdered and stored at 4°C for further analysis.

**Preparation of *Ceiba pentandra* leaf extract:** 10 g of shade dried *Ceiba pentandra* leaves were added to 100 ml of distilled water and boiled at 60°C for 15 min in water bath. Then, cool it down to room temperature and the solution was filtered through Whatmann no. 1 filter paper.

**Green synthesis of CP-CuNPs:** CP-CuNPs were synthesized by adding 25 ml of leaf extract to 250 ml 1M copper acetate solution and continuously stirred for one hour. Add a little drop of 1M NaOH and stir continuously for half an hour. After adding NaOH, the color change of solution from light brown to dark brown was observed. The CuNPs solution was centrifuged at 12000 rpm at 37 °C. Supernatant was removed and black precipitate that was obtained was filtered, washed and dried and stored for further use.

**Characterization of CP-CuNPs:** Synthesized CP-CuNPs were characterized by various methods including UV-visible spectrophotometry, SEM-EDAX, XRD, *in vitro* antioxidant activity.

**UV-Visible Spectroscopy:** CP-CuNPs copper nanoparticles were initially observed through the color change of the solution. The visual observation of color change from light brown to dark brown indicates the formation of copper nanoparticles. The copper nanoparticles formation was observed by measuring optical density (OD) in the various spectral ranges of 200-700 nm.

**SEM and EDAX analysis:** Thin films of CP-CuNPs were placed on a gold coated grid by just dropping a very small amount of sample on the grid. Extra sample was removed and thin films on SEM grid were allowed to dry by putting it under mercury lamp for 5 min and later examined at different angles. SEM (HITACHI Model SEM-JEOL) analysis was done at 120 kv. EDAX spectral analysis was observed at an acceleration voltage of 20-40 keV.

**XRD:** X-ray diffractometer (RIGAKU, SMARTLAB) operated at 30 Kv 100 Ma with Cu K $\beta$  as a radiation source was used to record XRD patterns of dried CP-CuNPs to confirm the crystalline nature.

#### **In vitro anti-oxidant assays**

**DPPH free radical scavenging assay:** The biosynthesized CP-CuNPs and leaf extract were tested for their antioxidant activity by DPPH method. Different concentrations of CP-CuNPs and leaf extract (50, 100, 150, 200 and 250  $\mu$ g/ml) were mixed with 3 mL of methanolic solution containing DPPH radicals (0.1 mM). Ascorbic acid was taken as positive control. After 30 min, absorbance was determined at 517 nm and converted into percentage by the given formula:

$$\% \text{ of inhibition} = [A_0 - A_1 / A_0] \times 100$$

where  $A_0$  = Absorbance of control and  $A_1$  = Absorbance of test.

**Hydrogen peroxide scavenging assay:** A modified protocol of Pavithra et al<sup>18</sup> was used for H<sub>2</sub>O<sub>2</sub> scavenging assay. Different concentrations of 50, 100, 150, 200 and 250  $\mu$ g/ml leaf extract of CP-CuNPs were mixed with 40 mM H<sub>2</sub>O<sub>2</sub> solution and incubated at room temperature for 10 min. The absorbance was measured at 230 nm and the percentage of H<sub>2</sub>O<sub>2</sub> scavenging by leaf extract, CP-CuNPs was monitored. Ascorbic acid was taken as positive control.

$$\% \text{ of scavenging (H}_2\text{O}_2) = [A_0 - A_1 / A_0] \times 100$$

where  $A_0$  = Absorbance of control  $A_1$  = Absorbance of test.

**Phosphomolybdenum assay:** A modified protocol of Pavithra et al<sup>18</sup> was used to estimate the total antioxidant activity. Molybdate reagent solution was prepared by 0.6 M sulphuric acid, 28 mM sodium phosphate and 4 mM ammonium molybdate. Different concentrations of 50, 100, 150, 200, 250  $\mu$ g/ml leaf extract of CP-CuNPs were added to each test tube individually containing 3 ml of distilled water and 1 ml of molybdate reagent solution followed by

incubation in water bath at 95 °C for 90 min. After incubation, these test tubes were cooled down to room temperature for 20-30 min and the absorbance was measured at 695 nm with ascorbic acid as positive control.

#### **Results and Discussion**

Synthesis of CP-CuNPs was confirmed due to colour change from blue to dark-brown. This was mediated by *Ceibapentandra* leaf extract which is involved in the reduction of Cu<sup>+</sup> to Cu. UV-Visible spectral analysis revealed a strong peak at 310 nm due to surface plasmon resonance (Figure 2). The above results are consistent with Murthy et al<sup>15</sup>, Ghosh et al<sup>10</sup> and Mali et al.<sup>13</sup> Secondary metabolites like phenolics, tannins, terpenoids etc. may act as a stabilizing and capping agent and may be responsible for the reduction of Cu<sup>+</sup> to Cu<sup>0</sup>.



**Figure 1: *Ceiba pentandra* plant**

SEM micrograph depicted in figure 3(A) showed that CP-CuNPs are spindle like structures and with average size of 62.63 nm. EDX spectrum displayed in figure 3(B) exhibited strong signal at the energy of 3 keV indicating the presence of CP-CuNPs and few signals of Si, C, O and Cl were also obtained.<sup>4,21</sup>

The crystalline nature of synthesized CP-CuNPs is presented in figure 4(A) which shows Bragg's reflections with  $2\theta$  range of 20-80 values corresponding to indexed planes of crystals of CP-CuNPs. The sharp peaks of the XRD pattern as in figure 4 indicated the crystalline nature of CuNPs<sup>4</sup>.

#### **In vitro antioxidant assays**

**DPPH assay:** *C. pentandra* leaf extract and CP-CuNPs were subjected to DPPH assay. CP-CuNPs showed the elevated % of DPPH free radical scavenging activity compared to *C. pentandra* leaf extract. Increasing concentration of CP-CuNPs increases the percentage of free radical scavenging activity. At 250  $\mu$ g/ml of concentration, CP-CuNPs show highest % of inhibition of about 80 % (Figure 5A)<sup>5</sup>.

**Phosphomolybdate method:** *C. pentandra* leaf extract and CP-CuNPs were subjected to determine the total antioxidant activity by phosphomolybdenum method.

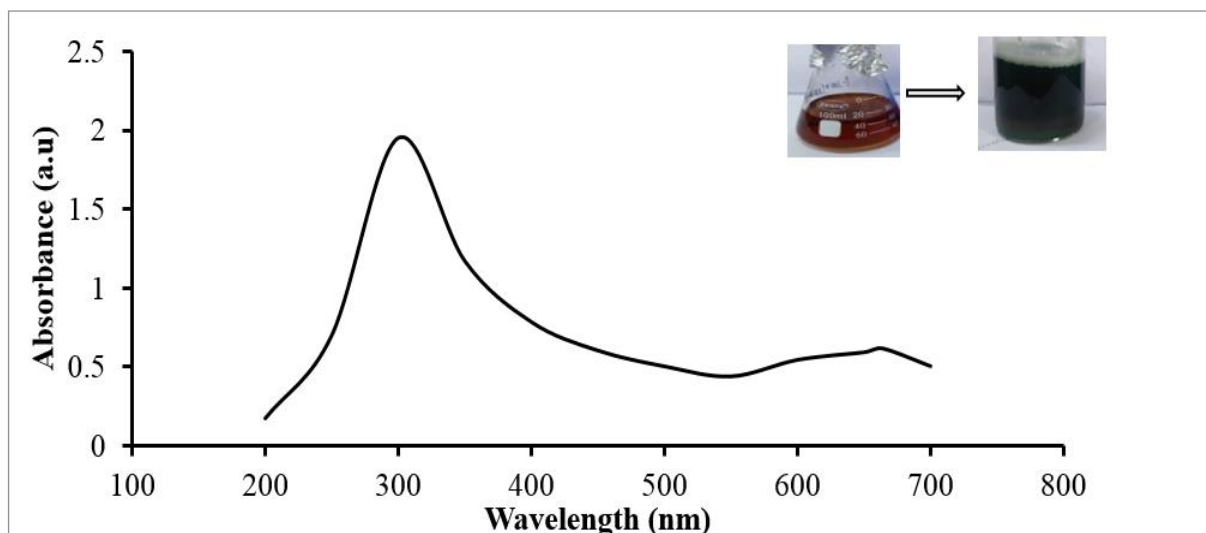


Figure 2: UV visible spectrum of CP-CuNPs. Colour change of *C. pentandra* leaf extract from deep brown to dark greenish-brown

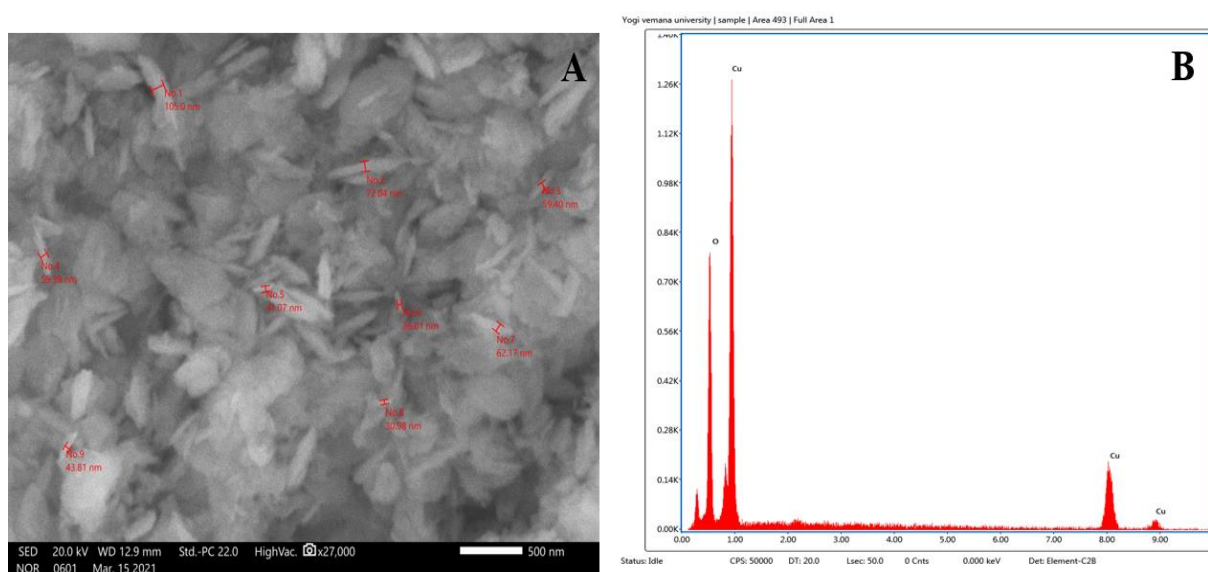


Figure 3: (A) SEM image and (B) EDAX image of CP-CuNPs

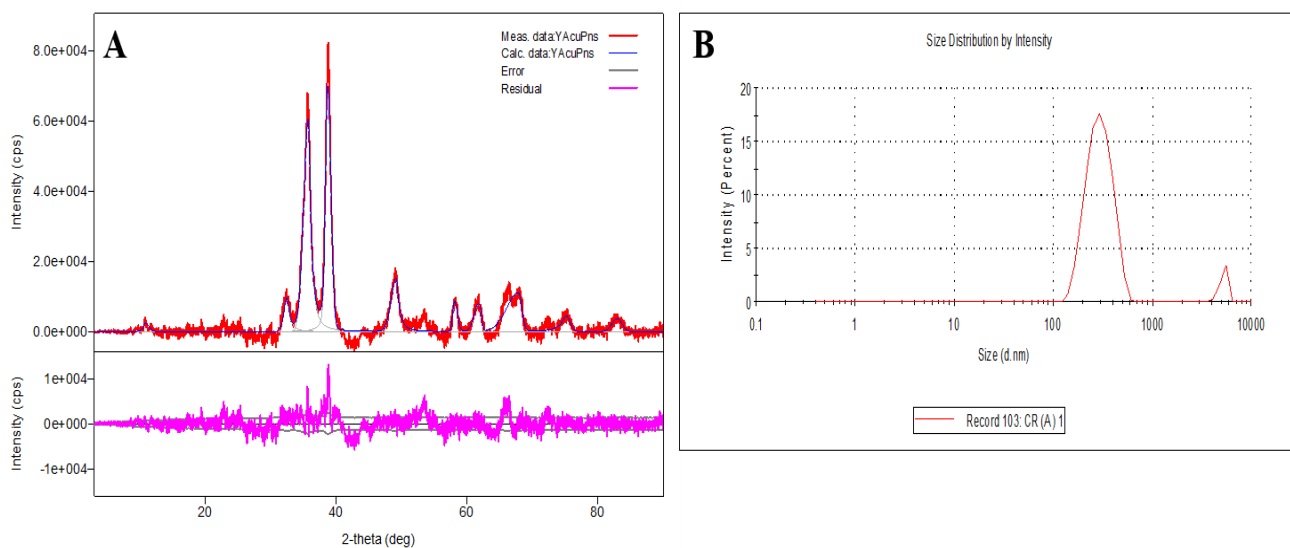
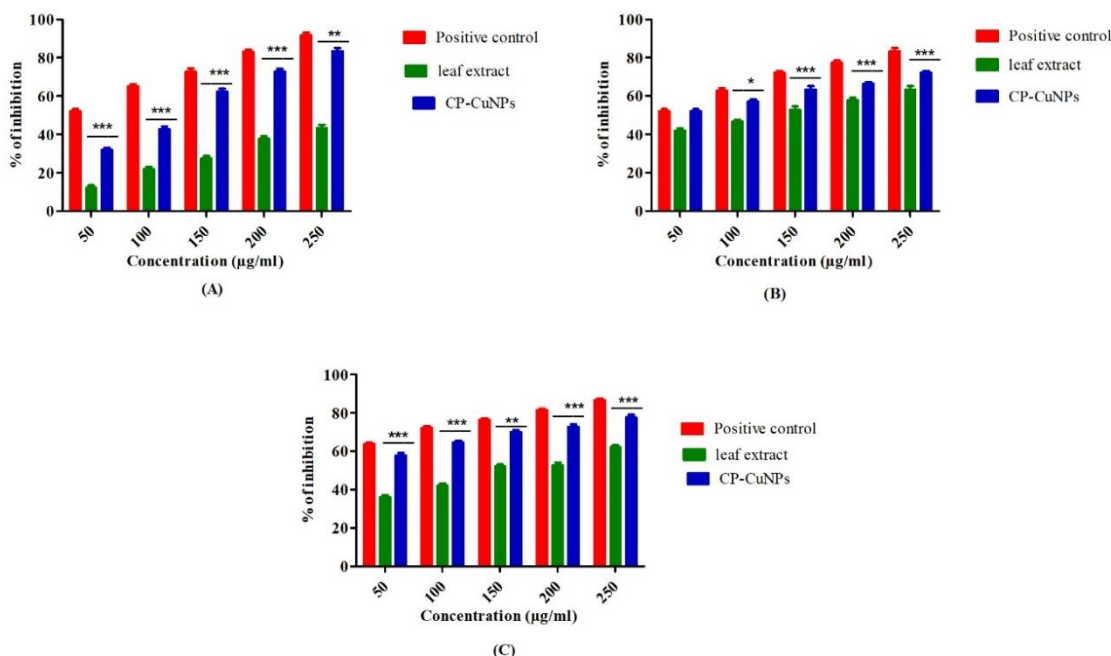


Figure 4: (A) XRD pattern and (B) particle size distribution of CP-CuNPs



**Figure 5: Antioxidant activity of *C. pentandra* leaf extract, CP-CuNPs and ascorbic acid as positive control.**  
**(A) DPPH Scavenging activity (B) Hydrogen peroxide scavenging activity**  
**(C) Total antioxidant activity by phosphomolybdenum method.**

All experiments were run in triplicate and values are expressed as mean  $\pm$  SEM. \*, \*\*, \*\*\* indicate the significance of CP-CuNPs in comparison with *C. pentandra* leaf extract

CP-CuNPs showed the higher % of total antioxidants activity compared to *C. pentandra* leaf extract. Increasing concentration of CP-CuNPs increases the % of total antioxidants activity. The % of total antioxidant activity of CP-CuNPs ranges from  $58\% \pm 0.55$  to  $80\% \pm 0.79$ . At 250 µg/ml of concentration, CP-CuNPs shows highest % of inhibition of about  $80\% \pm 0.79$  (Figure 5B) <sup>12</sup>. The basic principle behind this method is the subsequent conversion of Mo (IV) to Mo (V) showing maximum absorbance at 695 nm, used to determine the total antioxidant activity of the extracts. The more is the antioxidant activity, the higher is the absorption.

**Hydrogen peroxide scavenging assay:** *C. pentandra* plant extract and CP-CuNPs were subjected to H<sub>2</sub>O<sub>2</sub> free radical scavenging. CP-CuNPs showed the elevated % of H<sub>2</sub>O<sub>2</sub> free radical scavenging activity compared to *C. pentandra* leaf extract. Increasing concentration of CP-CuNPs increases the percentage of free radical scavenging activity. At 250 µg/ml of concentration, CP-CuNPs showed highest % of inhibition of about 75 % (Figure 5C) <sup>16</sup>.

## Conclusion

The study concludes that CP-CuNPs could be prepared instantly by making use of aqueous leaf extract of *C. pentandra*. The phytoconstituents such as alkaloids and phenolic compounds present in the leaf infusion may be responsible and act as both capping and reducing agent for synthesis of biogenic CuNPs as evident from EDX studies. The biosynthesized CP-CuNPs were found to have crystalline structure with face centered cubic (FCC)

geometry as obtained from XRD analysis. The SEM analysis confirmed the spindly shape of CP-CuNPs with average size of 62.63nm. The as-synthesized CP-CuNPs presented highest *in vitro* antioxidant activity revealing that these CP-CuNPs may be further evaluated for anticancer activity.

## References

- Ahmed S., Ahmad M., Swami B.L. and Ikram S., A review on plants extract mediated synthesis of silver nanoparticles for antimicrobial applications: a green expertise, *J. Advan. Res.*, **7(1)**, 17-28 (2016)
- Ahmed S. and Ikram S., Chitosan and its derivatives: a review in recent innovations, *Int. J. Pharmac. Scis. Res.*, **6(1)**, 14e30 (2015)
- Ameena S., Rajesh N., Syeda M. Anjum, Habeeb Khadri, Riazunnisa K., Arifullah M. and Abdul Kari Z., Antioxidant, Antibacterial and Anti-diabetic activity of green synthesized copper nanoparticles of *Cocculus hirsutus* (Menispermaceae), *Appl Biochem Biotech.*, <https://doi.org/10.1007/s12010-022-03899-4> (2022)
- Amer M.W. and Awwad A.M., Green synthesis of copper nanoparticles by *Citruslimon* fruits extract, characterization and antibacterial activity, *Chem Int*, **7(1)**, 1-8 (2021)
- Bello O.M., Oguntoye O.S., Adewumi Oluwasogo D.A., Bello O.E., Tijjani A.L.I., Alhaji A.A. and Adeniyi O., Phytobiological-facilitated production of silver nanoparticles from selected non-cultivated vegetables in Nigeria and their biological potential, *Turkish Journal of Pharmaceutical Sciences*, **17(6)**, 599 (2020)

6. Bindhu M.R. and Umadevi M., Antibacterial and catalytic activities of green synthesized silver nanoparticles, *Spectrochim. Acta A: Mol. Biomol. Spectrosc.*, **135**, 373-378 (2015)
7. Cady N.C., Behnke J.L. and Strickland A.D., Copper-based nanostructured coatings on natural cellulose: nanocomposites exhibiting rapid and efficient inhibition of a multidrug resistant wound pathogen, *A. baumannii* and mammalian cell biocompatibility *in vitro*, *Adv. Funct. Mater.*, **21**, 2506–2514 (2011)
8. Chatterjee A.K., Sarkar R.K., Chattopadhyay A.P., Aich P., Chakraborty R. and Basu T., A simple robust method for synthesis of metallic copper nanoparticles of high antibacterial potency against *E. coli*, *Nanotechn.*, **23**, 85–103 (2012)
9. Eltahir Mohamed Abdalla Elhadi and Umar Lawal Dano, Desertification Assessment and Mapping in Northern Shaanxi Province: A GIS-based and Remote Sensing Approach, *Res. J. Chem. Environ.*, **13**(2), 1-10 (2020)
10. Ghosh M.K., Sahu S., Gupta I. and Ghorai T.K., Green synthesis of copper nanoparticles from an extract of *Jatropha curcas* leaves: Characterization, optical properties, CT-DNA binding and photocatalytic activity, *RSC Advances*, **10**(37), 22027-22035 (2020)
11. Grass G., Rensing C. and Solioz M., Metallic copper as an antimicrobial surface, *Appl. Environ. Microbiol.*, **77**(5), 1541–1547 (2011)
12. Loganayaki N., Siddhuraju P. and Manian S., Antioxidant activity and free radical scavenging capacity of phenolic extracts from *Helicteres isora* L. and *Ceiba pentandra* L., *Journal of Food Science and Technology*, **50**(4), 687-695 (2013)
13. Mali S.C., Dhaka A., Githala C.K. and Trivedi R., Green synthesis of copper nanoparticles using *Celastrus paniculatus* Willd. leaf extract and their photocatalytic and antifungal properties, *Biotechnology Reports*, **27**, e00518 (2020)
14. Mikolay A., Huggett S., Tikana L., Grass G., Braun J. and Nies D.H., Survival of bacteria on metallic copper surfaces in a hospital trial, *Appl. Microbiol. Biotech.*, **87**, 1875–1879 (2010)
15. Murthy H.C., Desalegn T., Kassa M., Abebe B. and Assefa T., Synthesis of green copper nanoparticles using medicinal plant *Hagenia abyssinica* (Brace) JF. Gmel. leaf extract: Antimicrobial properties, *J. Nanomat.*, <https://doi.org/10.1155/2020/3924081> (2020)
16. Nguetefack T.B., Fofie C.K., Nguetefack-Mbuyo E.P. and Wuyt A.K., Multimodal  $\alpha$ -glucosidase and  $\alpha$ -amylase inhibition and antioxidant effect of the aqueous and methanol extracts from the trunk bark of *Ceiba pentandra*, *Bio Med Research International*, <https://doi.org/10.1155/2020/3063674> (2020)
17. Padalia H., Moteriya P. and Chanda S., Green synthesis of silver nanoparticles from marigold flower and its synergistic antimicrobial potential, *Arab. J. Chem.*, **8**(5), 732-41 (2015)
18. Pavithra K. and Vadivukkarasi S., Evaluation of free radical scavenging activity of various extracts of leaves from *Kedrostis foetidissima* (Jacq.) Cogn, *Food Science and Human Wellness*, **4**(1), 42-46 (2015)
19. Pourmortazavi S.M., Taghdiri M., Makari V. and Rahimi-Nasrabadi M., Procedure optimization for green synthesis of silver nanoparticles by aqueous extract of *Eucalyptus oleosa*, *Spectrochim. Acta A: Mol. Biomol. Spectrosc.*, **136**, 1249-1254 (2015)
20. Raffi M., Mehrwan S., Bhatti T.M., Akhter J.I., Hameed A., Yawar W. and Hasan M.M., Investigations into the antibacterial behavior of copper nanoparticles against *Escherichia coli*, *Ann. Microbiol.*, **60**, 75–80 (2010)
21. Rajesh K.M., Ajitha B., Reddy Y.A.K., Suneetha Y. and Reddy P.S., Assisted green synthesis of copper nanoparticles using *Syzygium aromaticum* bud extract: Physical, optical and antimicrobial properties, *Optik*, **154**, 593-600 (2018)
22. Santo C.E., Quaranta D. and Grass G., Antimicrobial metallic copper surfaces kill *Staphylococcus haemolyticus* via membrane damage, *Microbiology Open*, **1**(1), 46–52 (2012)
23. Sre P.R.R., Reka M., Poovazhagi R., Kumar M.A. and Murugesan K., Antibacterial and cytotoxic effect of biologically synthesized silver nanoparticles using aqueous root extract of *Erythrina indica* lam, *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, **135**, 1137e1144 (2015)
24. Srivastava A., Antiviral activity of copper complexes of isoniazid against RNA tumor viruses, *Resonance*, **14**(8), 754–760 (2009)
25. Tagad C.K. et al, Green synthesis of silver nanoparticles and their application for the development of optical fiber based hydrogen peroxide sensor, *Sensors and Actuators B*, **183**, 144– 149 (2013)
26. Thuesombat P., Hannongbua S., Akasit S. and Chadchawan S., Effect of silver nanoparticles on rice (*Oryza sativa* L. cv. KDML 105) seed germination and seedling growth, *Ecotoxic. Environ. Safety*, **104**, 302-309 (2014)
27. Tian K., Cailin Liu, Haijun Yang and Xianyan Ren, *In situ* synthesis of copper nanoparticles/polystyrene composite, *Colloids and Surfaces A: Physicochem. Eng. Aspects*, **397**, 12– 15 (2012)
28. Yamini Sudha Lakshmi G. et al, Green Synthesis of Silver Nanoparticles from *Cleome viscosa*: Synthesis and Antimicrobial Activity, *IPCBE*, **5**, 334-337 (2011).

(Received 24<sup>th</sup> March 2022, accepted 26<sup>th</sup> May 2022)